found

# Batch | offline ML

the system collect data over a period of time and process it all at once in a large group or batch.

$\boxed{\text{Data}} \rightarrow \text{model} \rightarrow \boxed{\text{test}} \rightarrow \text{serve}$

## Problem with Batch learning

New Data → Old data → model → test → Serve

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daily or weekly updates

- Lot of Data
- Hardware Limitation
- Availability

- Online ML
  - also called incremental learning
  - model constantly updates as new data arrives. It doesn't wait for batch, it learn from data point one by one

When to use?

- constant changing environment like stock market.
- news.
- cost Effective
- Faster solution.

## Instance Based Learning

System learn training example by Heart and then generalize to new instances based on some similarity measure. It is called instance-based because it builds the hypothesis from training instances.

It is also known as memory-based learning.

- Time complexity of this algorithm depends upon size of training data. Worst-case time complexity of this algo. is O(n) where n is number of ~~tim~~ training instances.

Model-based learning

A system is called model-based when it learn from the data and create a model which has some parameter and it predicts the output by using this training data.

# Challenges in ML

1. Data collection
2. Insufficient Data / Labelled Data
3. Non-Represenfive Data
4. Poor Quality Data
5. Irrelevant Feature / column
6. Overfitting
7. Underfitting
8. Software Integration
9. Offline learning / Deployment
10. Cost Involved

# MLDLC

1. Framing the Problem
2. Data Preparation
3. Data Wrangling
4. Analyse Data
5. Train model
6. Test model
7. Deploy model
8. optimize

Frame Problem
Gathering Data
Data PreProcessing
EDA
Feature Engineering
Model Training, Evaluation
Model Deployment
Testing
Optimize